



First-Time Wi-Fi Connection

Put your CarthaBot on your home Wi-Fi — once — using the "First time? Put the robot on your Wi-Fi" button.

CarthaBot Companion · in-app setup over the USB cable

The idea: the robot and your PC join the **same Wi-Fi**, then the PC sends programs to the robot over that network. You set this up **once over the USB cable** — after that it's wireless on your own network, no switching, internet kept.



Before you start — 3 things that MUST be true

If the robot "doesn't join," it's almost always one of these. Check them first:

- 1 The bridge firmware v4 is flashed on the ESP32-C3.** Without it, the Save does nothing. Flash `Firmware\esp32c3_bridge_v4_appwifi\esp32c3_bridge_v4_appwifi.ino` in Arduino IDE once. (See "How to check" at the end — this is the #1 cause.)
- 2 Your Wi-Fi is 2.4 GHz.** The ESP32-C3 has **no 5 GHz radio**. If your network is 5 GHz only, it can't join — use the 2.4 GHz one.
- 3 You type the Wi-Fi name EXACTLY.** It's case- and character-exact — e.g. **TT_2560** (underscore), not `TT-2560` (hyphen).

Step by step — first-time connection

- 1 Plug the CarthaBot into the PC** with the USB-C cable and turn it on. (The Save sends your Wi-Fi to the robot through this cable.)
- Open the **CarthaBot Companion** app and open any coding activity (Kids Coding / under-6 / Advanced Python).
- In the connection panel, on the **WiFi** card, click **First time? Put the robot on your Wi-Fi**.
- In **Your Wi-Fi name (SSID)**, type your network name **exactly** (e.g. `TT_2560`).
- In **Wi-Fi password**, type your Wi-Fi password (special characters like `@` are fine).
- Click **Save to robot (USB cable)**.
- Wait about 15 seconds — don't unplug yet.** The app sends the Wi-Fi to the robot, then checks your network to see if it joined, and shows the result:
 - ✓ **"CarthaBot joined your Wi-Fi! It's at 192.168.1.xx"** → success! That number is the robot's address.

- ⚠️ **"CarthaBot didn't appear on your Wi-Fi..."** → it didn't join — re-check the 3 things above, then Save again.

8 On success: click **Close, unplug the USB cable**, and press **Connect**. The app finds the robot on your network automatically and you're ready to ► **Play**.

Done once, done forever. The robot remembers your Wi-Fi and rejoins it every time it powers on. From now on: power the robot, open the app, press **Connect**, and code — no cables, no network switching, internet kept.

What the messages mean

Message	Meaning / what to do
"Looking for the robot on USB..."	The app is finding the robot's USB port. Make sure the cable is plugged in and the robot is on.
"Couldn't find the robot on USB"	The robot isn't on USB. Plug the cable into the robot and the PC, turn it on, try again.
"Saved! Waiting for CarthaBot to join..."	Good — the Wi-Fi was sent. Wait ~15 s while it joins and the app verifies.
✅ "CarthaBot joined your Wi-Fi! It's at ..."	Success. Unplug USB and press Connect.
⚠️ "didn't appear on your Wi-Fi"	It didn't join — wrong name/password, or a 5 GHz network, or v4 not flashed. Fix and Save again.
"The robot reported an error..."	MicroPython isn't set up. Open a coding activity over USB once (the app installs it), then Save again.

If it won't join — the definitive check

To see exactly what the robot's Wi-Fi chip is doing, read its own report:

- 1 Plug the **Beetle ESP32-C3** into the PC by **its own USB-C** port.
- 2 Arduino IDE → **Tools** ▶ **Serial Monitor**, baud **115200**. Press the Beetle's **RESET**.
- 3 Read the line that starts with `[wifi]` :
 - `STA=on 192.168.1.xx` → it joined ✅ (use that IP in the app).
 - `STA=off` → it tried but failed → wrong password / 5 GHz / out of range.
 - **No STA line at all** → the `v4` firmware isn't flashed → flash it (prerequisite #1).

Troubleshooting

Problem	Fix
"didn't appear" after Save	1) Flash <code>v4</code> if not done. 2) Use the 2.4 GHz network. 3) Type the SSID exactly (<code>TT_2560</code> , underscore). 4) Re-check the password. Then Save again.
It joined but Connect can't reach it	Make sure the PC is on the same Wi-Fi as the robot. Press Connect again (it auto-scans). Or type the robot's IP (from the ✅ message) in the WiFi address box.

Want to change to a different Wi-Fi later	Repeat this whole flow over USB with the new network name + password.
No internet while setting up	Not an issue here — this setup is over the USB cable, so your PC stays on your normal Wi-Fi the whole time.

CarthaBot Companion — first-time Wi-Fi setup is one-time per robot per network, done over USB (no network switching). The ESP32-C3 is 2.4 GHz only. Requires the **v4** bridge firmware on the ESP32-C3 and MicroPython on the robot (the app installs MicroPython's boot helper automatically over USB).